Predicting SpaceX Falcon 9 First-Stage Landing Success

Applied Data Science Capstone Project Report

Project objective

Use launch history and machine-learning classification to investigate whether a Falcon 9 first stage will land successfully. A reusable first stage can reduce launch costs, so understanding factors associated with landing outcomes is valuable for launch planning.

Prepared for: IBM Applied Data Science Capstone

Author: [Enter your name]

Important integrity note: this report is a structured presentation draft. Replace any project-specific results and the repository placeholder with outputs from your course labs before submitting. Do not present uncomputed model scores as measured results.

1. Executive Summary

Problem: estimate the likelihood of a successful Falcon 9 first-stage landing using historical launch information.

Approach: collect launch records, clean and transform variables, explore patterns with plots and SQL, map launch sites, and compare classification models.

Expected deliverables: reproducible analysis, clear visualizations, a model evaluation, and practical recommendations.

Key takeaway: landing outcomes should be evaluated using multiple launch attributes rather than assuming one factor determines success. Final conclusions must be tied to the computed results from the notebooks.

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2. Introduction and Problem Statement

Falcon 9 is a partially reusable launch vehicle. Recovering and reusing its first stage can reduce the cost of access to space.

Business question: can historical launch data help predict first-stage landing success and identify factors associated with the outcome?

Target variable: landing outcome, encoded as a binary label (1 = successful landing; 0 = unsuccessful landing), following the course dataset definition.

Scope: analyze historical launches and compare predictive models. The model supports analysis; it does not guarantee the outcome of a future launch.

3. Data Collection — SpaceX API

Purpose: retrieve structured launch records from the SpaceX API, including launch dates, rocket details, launch sites, payload information, and outcomes where available.

Workflow

1. Send an HTTP GET request to the relevant SpaceX API endpoint.
2. Check the response status and parse the JSON payload.
3. Select fields relevant to the prediction task.
4. Convert nested JSON fields into tabular columns.
5. Save the extracted records for repeatable analysis.

API endpoint used in the course workflow: https://api.spacexdata.com/v4/launches/past

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4. Data Collection — Web Scraping

Purpose: obtain launch-site or mission details from a public webpage when those details are not readily available in the selected API response.

Workflow

1. Request the relevant webpage.
2. Parse the HTML with a tool such as BeautifulSoup.
3. Locate the target table or elements by their HTML tags and attributes.
4. Extract rows and columns into a structured table.
5. Validate the extracted values and merge them using a consistent key.

Use scraping responsibly: respect site terms, robots guidance, rate limits, and attribution requirements. Confirm that scraped fields align with the API data before merging.

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5. Data Wrangling Methodology

Data wrangling prepares raw records for trustworthy analysis and modelling.

Cleaning steps

- Inspect column names, data types, duplicate rows, and missing values.
- Keep only fields relevant to the business question.
- Convert dates and numeric fields to appropriate data types.
- Handle missing payload mass or categorical values using a documented rule.
- Create the binary landing-success label according to the project definition.
- Encode categorical features and split training/testing data before model evaluation.

Quality checks: verify row counts before and after cleaning, inspect label balance, and ensure that target information does not leak into the predictor features.

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6. EDA with Data Visualization

Exploratory data analysis (EDA) is used to understand distributions, compare groups, identify relationships, and detect data-quality issues before modelling.

Recommended plots and the question each answers

- Scatter plot: payload mass versus launch outcome — does outcome vary with payload mass?
- Bar chart: success counts by launch site — are outcomes distributed differently across sites?
- Yearly trend: launch counts or success rate over time — how has the launch programme changed?
- Box plot / histogram: payload mass distribution — are values skewed or extreme?

Interpret only what the plotted data supports. A visible association is not proof that one variable causes success.

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7. EDA with SQL

SQL provides a reproducible way to summarize the cleaned launch table and answer targeted questions.

Example query patterns

```
SELECT LaunchSite, COUNT(*) AS launches
FROM SPACEXTBL
GROUP BY LaunchSite
ORDER BY launches DESC;
```

```
SELECT BoosterVersion, AVG(MissionOutcome) AS success_rate
FROM SPACEXTBL
GROUP BY BoosterVersion
ORDER BY success_rate DESC;
```

Additional analyses: payload ranges, launch-site rankings, success rates, and launch-date/time patterns. Adapt table and column names to the actual course database and report the returned values from your run.

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8. Interactive Visual Analytics

Folium launch-site map

Plot launch-site markers on an interactive map. Include a popup with the site name and relevant launch records. Use marker clusters if points overlap.

Proximity analysis

Compare each launch site with nearby coastlines, roads, or populated areas using geographic coordinates and an appropriate distance measure. Explain why proximity and geography matter for launch operations.

Plotly Dash app

A simple dashboard can include a success/failure pie chart and a payload-mass-versus-outcome scatter plot, with filters for launch site, year, and booster version.

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9. EDA Results — Launch Sites and Trends

Insert the actual outputs from the course notebooks in this section.

Required result slides should cover:

- Launch site markers and launch records on the Folium map.
- Launch-site counts and success-rate comparisons.
- Payload-versus-outcome scatter plot.
- Bar charts and yearly launch trends.
- SQL results for site, payload, success rates, rankings, and time analysis.

Interpretation template: “The chart shows [observed pattern]. This matters because [operational implication]. The conclusion is limited by [sample size / missing data / other limitation].”

Do not leave this section as a substitute for required notebook result screenshots if your rubric expects actual outputs.

10. Predictive Analysis — Modelling Plan

Task type: binary classification — predict successful versus unsuccessful landing.

Models to compare

- Logistic Regression: interpretable baseline.
- Support Vector Machine (SVM): learns a separating boundary.
- Decision Tree: captures nonlinear rules and is relatively interpretable.
- K-Nearest Neighbors (KNN): predicts from nearby training examples.

Process: split the data into training and test sets; fit preprocessing only on training data; tune models with cross-validation when possible; evaluate each model on the same held-out test set.

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11. Model Evaluation and Confusion Matrix

Report model performance from your actual run; do not invent scores.

Metrics

- Accuracy: fraction of all predictions that are correct.
- Precision: fraction of predicted successes that were actually successes.
- Recall: fraction of actual successes correctly identified.
- F1 score: harmonic mean of precision and recall.
- Confusion matrix: counts true positives, true negatives, false positives, and false negatives.

A confusion matrix makes error types visible. If the classes are imbalanced, accuracy alone can be misleading; report additional metrics and the class distribution.

Insert: model comparison table, confusion matrix for the selected model, and the course notebook's evaluation outputs.

12. Best Model and Interpretation

Choose the best model based on the held-out evaluation and the project objective, not simply on training accuracy.

Model selection checklist

1. Compare the same metrics across all candidate models.
2. Review the confusion matrix and the errors most relevant to the task.
3. Check whether performance is stable and whether data leakage is possible.
4. Explain the selected model in plain language.
5. State limitations and avoid claiming certainty for individual future launches.

Final statement template: “The [model name] performed best on [metric] in the test evaluation. Its main limitation is [limitation]. Further validation on newer launches is recommended.”

13. Conclusions and Recommendations

This project demonstrates an end-to-end data science workflow: data collection, cleaning, exploration, SQL analysis, interactive visualization, and predictive modelling.

Recommendations

- Keep the data pipeline reproducible and document any changes to source data.
- Monitor model performance as new launches become available.
- Use class-aware metrics and inspect errors before operational use.
- Treat model predictions as decision support, not as a guarantee of landing success.
- Re-run analysis when booster versions, payload profiles, or operational conditions change.

Replace this general conclusion with 2–3 specific findings supported by your notebook outputs.

14. References, Reproducibility and Submission Checklist

Sources to cite in the final version

- SpaceX API: https://api.spacexdata.com/v4/launches/past
- IBM Applied Data Science Capstone course materials and the project notebooks you completed.
- Any external webpage used for scraping or launch-site information.

Before submitting

- ■ Replace every [PASTE YOUR ACTUAL PROJECT GITHUB URL HERE] placeholder.
- ■ Add your name and any required course details.
- ■ Add actual charts, SQL outputs, map screenshots, dashboard screenshots, and model evaluation results from your labs.
- ■ Remove this checklist and any instructions that are not part of your final presentation.
- ■ Export the completed report to PDF and upload it as Data Science Capstone Project Report.

This is a presentation framework, not evidence that the lab analyses or models have already been run.

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